



Digital Transformations for  
Supporting Next-Generation Labour

Co-funded by the  
Erasmus+ Programme  
of the European Union



# MetaLab

MacPaw AI Lab in VRChat for the NEXT-Study Metaverse

## Nomination I — Metaverse representation of the university

### Team MetaLab

Igor Sikorsky Kyiv Polytechnic Institute

### Team members

Sviatoslav Pavlenko, Denys Iliencko, Dmytro Poshytyniuk,  
Mark Mamenko, Kateryna Shozda

[nextstudy.eu](https://nextstudy.eu)

## **Sviatoslav Pavlenko**

Igor Sikorsky KPI

### **Role in the project**

Team Lead. Coordination, task split, deadlines and communication with the consultant.

## **Denys Iliencko**

Igor Sikorsky KPI

### **Role in the project**

3D Artist. Room and equipment modelling, retopology, mesh prep for texturing.

## **Dmytro Poshytyniuk**

Igor Sikorsky KPI

### **Role in the project**

Look Development. PBR textures and Unity materials for realistic surfaces.

## **Mark Mamenko**

Igor Sikorsky KPI

### **Role in the project**

UdonSharp Developer. Interactive terminals, object state sync across players.

## **Kateryna Shozda**

Igor Sikorsky KPI

### **Role in the project**

QA / VR Tester. In-headset checks, FPS, collisions and network stability.

## **How we work**

Agile team of five. Modelling, texturing, C# logic and Unity assembly run in parallel, synced over Git. Regular VR playtests.

## ANNOTATION

### Digital twin of our lab in VRChat

MacPaw AI Lab · NEXT-Study Metaverse

For the contest we published a VRChat rebuild of KPI MacPaw AI Lab: seminar rows, lounge, boards and workstations in one multiplayer hall. Built in Unity 2022.3.22f1 + VRChat SDK3 with UdonSharp, Blender retopo, Substance PBR and baked lighting.

## PROJECT GOALS

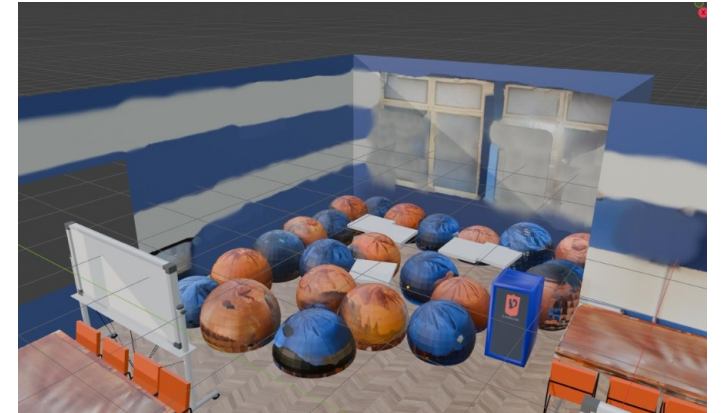
- Visual accuracy: real proportions of furniture and gear
- Functional space for SE and networking practice
- Extensible layout for later AI / simulators
- Stable FPS via retopo, PBR and light baking
- Multiplayer VRChat world for NEXT-Study

# Selected Location

NEXT

## MacPaw AI Lab

Photos below are our laboratory only — seminar, lounge and workstations. No stock or generated interiors.



# The space in VRChat

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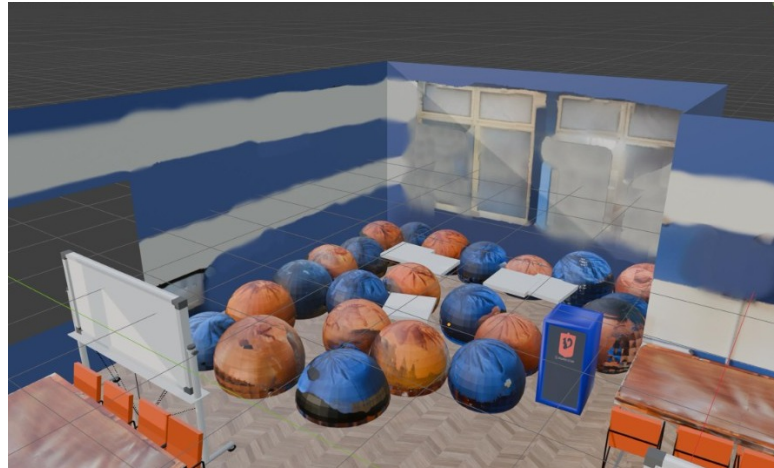


**Classroom layout**



Retopo'd mesh, baked lighting (no realtime lamps), collisions on floor, tables and walls.

**Lounge and boards**



Lounge and boards. Wall logo is a raster from our lab photos, not generated stock.

**Workstations**



Workstations and UdonSharp sync. Space reserved for the CodeSensei terminal.

## Tools and software

Unity (scene, collisions, light baking) · VRChat SDK3 / Udon · UdonSharp (C#) · Blender · Adobe Substance 3D Painter (PBR) · Git.

## Development process

1) Block out MacPaw AI Lab. 2) Model and retopo furniture and equipment. 3) Paint PBR materials. 4) Bake lighting and collisions in Unity. 5) Add UdonSharp interaction and publish the world.

## VRChat integration

SDK3 Worlds: spawns, audio zones, synced object state. Published world wrld\_b1c73436-f022-4f98-9172-671f9f0da989. Prefer PC / PCVR.

## Difficulties and limits

Polygon and material budgets. Baking instead of realtime lights. Headset FPS vs VR sickness. Udon networking. Quest only after a separate optimisation pass.

## **Educational use**

In NEXT-Study Metaverse a group meets in the same virtual hall for remote practicals, hackathons and applicant tours — with the spatial feel of a real classroom.

## **User experience**

Spawn, walk the aisles, sit at a workstation, use boards and equipment, see classmates in the same room. Presence replaces a flat video call.

## **Further development**

Hook AI mentors and network simulators through OSC and HTTP, refine Quest performance, keep adding equipment as modular prefabs.

# Demo and Project Materials

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## Project demo

[https://vrchat.com/home/world/wrld\\_b1c73436-f022-4f98-9172-671f9f0da989/info](https://vrchat.com/home/world/wrld_b1c73436-f022-4f98-9172-671f9f0da989/info)

## Project materials

GitHub: [github.com/mamenkomarkus-creator/MetaLab-NEXT](https://github.com/mamenkomarkus-creator/MetaLab-NEXT)

Docs: [presentation-materials/documentation/DOCUMENTATION.md](#)

World ID: wrld\_b1c73436-f022-4f98-9172-671f9f0da989

## Additional resources

VRChat: [hello.vrchat.com](https://hello.vrchat.com) NEXT: [nextstudy.eu](https://nextstudy.eu)



GitHub



VRChat



# NEXT

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Thank you!  
Questions?

